

POWERFACTORY

PowerFactory 2024

Technical Reference

DIgSILENT Photovoltaic System Templates

DIgSILENT PV System 1PH 50Hz,
DIgSILENT PV System 1PH 60Hz,
DIgSILENT PV System 1PH-N 50Hz,
DIgSILENT PV System 1PH-N 60Hz,
DIgSILENT PV System 3PH 50Hz,
DIgSILENT PV System 3PH 60Hz

PF2024

POWER SYSTEM SOLUTIONS
MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

Copyright © 2024 DIgSILENT GmbH
All rights reserved. No part of this publication may be reproduced or distributed in any form without written permission of DIgSILENT GmbH.

January 11, 2024
PowerFactory 2024
Revision 3

Contents

1 Introduction **1**

2 How to use the Templates **2**

3 Model Description **2**

4 How to Customize the Model **4**

4.1 Load Flow 4

4.2 Short Circuit (VDE / IEC / ANSI / complete) 4

4.3 Dynamic Simulation (RMS) 5

5 Using temperature and radiation profiles **7**

6 References **8**

List of Figures **9**

List of Tables **10**

1 Introduction

This document describes the following photovoltaic system templates:

- DIgSILENT PV System 1PH 50Hz
- DIgSILENT PV System 1PH 60Hz
- DIgSILENT PV System 1PH-N 50Hz
- DIgSILENT PV System 1PH-N 60Hz
- DIgSILENT PV System 3PH 50Hz
- DIgSILENT PV System 3PH 60Hz

The rated power reaches from 40 kVA (single phase) up to 500 kVA (three phase). The template is a generic model for a photovoltaic system. It represents the grid side converter and the photovoltaic cell (modelled in DSL). The PV system model can be used for

- Load Flow Studies
- Stability Studies (RMS simulation)
- The three-phase templates ("DIgSILENT PV System 3PH 50Hz", "DIgSILENT PV System 3PH 60Hz") are suitable for balanced and unbalanced RMS simulation, however the controls are only implemented for positive sequence. The three-phase templates can be used in EMT simulation for limited use cases, as long as only balanced grids and balanced events are simulated, because the measurement signals are not filtered from unbalances or noise.
- The single-phase templates ("DIgSILENT PV System 1PH 50Hz", "DIgSILENT PV System 1PH 60Hz", "DIgSILENT PV System 1PH-N 50Hz", "DIgSILENT PV System 1PH-N 60Hz") require always an unbalanced RMS simulation. The single-phase templates are not suitable for EMT simulation.

For dynamic simulation, a variable step size (automatic step size adaptation) is possible. The integration step size for (electromechanical) transients should be smaller than 5 ms (for fixed or variable step size). Higher values can be used for the maximum step size (automatic step size adaptation), depending on the network model, events and effects under investigation.

2 How to use the Templates

A template can be added to an active project as follows:

- Press the *General Templates* button in the drawing toolbox.
- A list of available templates will be shown. Select from the sub-folder “Library Templates”, “Photovoltaic” the one you want to use.
- Click into the drawing area. You can see the shape of the selected template attached to the mouse pointer.
- Place the template with another mouse click.
- Close the template window after you finished adding the PV models.

Do not delete the terminal to which the generator is connected - this would destroy the measurement points of all measurement devices (located in the corresponding composite model).

You can remove a placed model directly by pressing the undo button () or you can delete the model also manually. If you want to delete a model manually you should mind that the model consists of visible elements in the single line diagram and also of a composite model which is only visible in the Data Manager.

3 Model Description

The photovoltaic system is in the single line diagram represented by a PV system element [2]. The models of the controllers as well as the photovoltaic cell are collected in a composite model *Photovoltaic System 1/3PH*. This composite model for the control can be found either through the link *Plant Model* on the Basic Data page of the PV system or with the Data Manger in the used grid.

The composite model *Photovoltaic System 1/3PH* is created from the frame definition *Frame PV System*, which is shown in Figure 3.1.

The individual slots are briefly described in Table 3.1. Each slot can be filled with either a DSL-model, a measurement device or a *PowerFactory* element, such as a PV system. All measurement devices are connected either to the terminal or to the cubicle, which connects the generator with the terminal. This approach ensures that all measurement devices are correctly connected after entering the model using the template.

Table 3.1: Frame description

Slot Name	Description	Needed Type
Static Generator	Representation of the inverter in the single line diagram	*.*ElmGenstat* or *.*ElmPvsys*
Photovoltaic Model	DSL model of the PV cell	*.*ElmDsl*
DC Busbar and Capacitor Model	DSL representation of the DC system (is not included in the PV system model)	*.*ElmDsl*
Controller	Control unit, regulates the DC voltage and the reactive power	*.*ElmDsl*
Qreference	DSL model for the control modes of the reactive power supply	*.*ElmDsl*
Protection	DSL model of the protection device	*.*ElmDsl*
PowerMeasurement	Power measurement device, needed for control feedback	*.*StaPqmea*
AC Voltage	AC voltage measurement device, needed for fault detection	*.*StaVmea*
PhaseMeasurement	PLL, needed for voltage angle measurement, input to the PV system	*.*ElmPhi__pll*
ActivePowerReduction	DSL model with power reduction logic in case of over frequency	*.*ElmDsl*
Slow Frequency Measurement	PLL, with slow settings for measuring the frequency	*.*ElmPhi_pll*
Solar Radiation	Slot for radiation selection model.	*.*ElmDsl*
Temperature	Slot for temperature selection model	*.*ElmDsl*
Measurement Solar Radiation	Slot for Measurement file with radition profile	*.*EmlFile*
Measurement Temperature	Slot for Measurement file with temperature profile	*.*ElmFile*

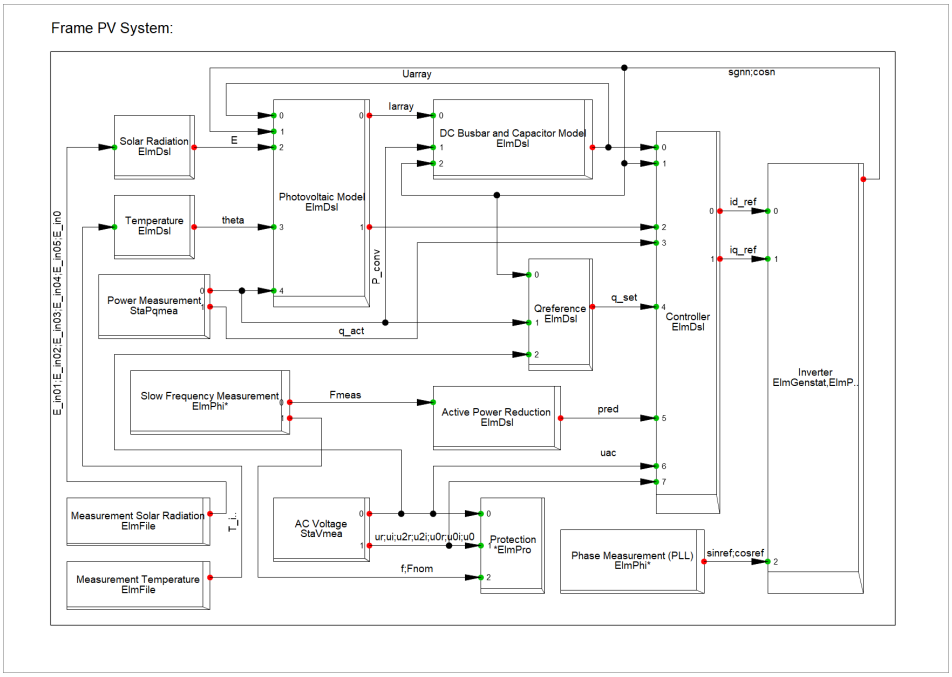


Figure 3.1: Frame Definition *Frame PV System* (*.BlkDef)

4 How to Customize the Model

You will find in this chapter a description how you can configure the model from the template according to your specific needs. The configuration is sorted to the calculation functions because there are fewer configurations needed for load flow calculation than for a dynamic simulation.

The model is generic and therefore completely open - you can configure it according to your needs. Following, you will find a short manual for changing some settings of the PV-system. The PV system consists in the single line diagram of a PV system element (the inverter) and of some DSL models, linked together in the composite model *Photovoltaic System 1/3PH*.

4.1 Load Flow

Changing the connection phase

This applies only to the single phase PV template. The PV system element is connected to a single phase terminal. This can be connected using a single phase breaker to any AC system terminal in *PowerFactory*. By default, the added breaker will connect the PV system to phase A. This can be changed by editing the breaker element. After pressing the “Figure” button the connection will be visible at the bottom of the window. The phase can then be changed by double clicking on the red cubicle name. This will open the edit dialogue of the cubicle. Here, the phase can be changed within the corresponding drop down menu.

Changing the rated (peak) power

The rated power is for the load flow model only defined by the PV system data. If you want to change the rated power you have to change this on the basic data page of the PV system (don’t forget to change also the power setpoint on the load flow page).

Changing the active power power output

The active power is for the load flow model only defined by the PV system data. If you want to change the active power you have to change this on the load flow page of the PV system (don’t forget to change also the rated power on the basic data page, point above).

Changing the nominal AC voltage

You can also change the nominal voltage of the converter if your PV-system has a different nominal grid side voltage. The only modification you have to do is changing the nominal voltage of the terminal to which the PV system is connected.

Note: There is no saturation considered. It is assumed that the DC voltage is all the time high enough.

4.2 Short Circuit (VDE / IEC / ANSI / complete)

The short circuit behaviour of the model is also completely determined by the element settings. Therefore, there are no changes in the dynamic model parts necessary. It is important to note that the IEC60909:2016 fault contribution is set by default to zero. The user needs to change these settings for his specific converter to get realistic results.

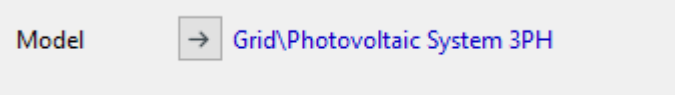
4.3 Dynamic Simulation (RMS)

The main purpose of the model added with the template is of course the dynamic simulation. The great advantage of using a template for creating the PV-model is that all measurement points are already correctly defined and all common models are already configured.

The model is generic and therefore completely open - you can configure it according to your needs. Following you will find a short manual for changing some settings of the PV-system.

Changing the rated (peak) power

The rated (peak) power is defined by the PV system data. If you want to change the rated power, you can do this on the basic data page of the PV system. The same applies to the rated power factor. In the *PV Array* model, the user can optionally change the number of modules in series (parameter *n_series*) and the oversizing factor of the plant. This factor indicates the ratio between installed module power and rated active power of the inverter(s). Based on the number of modules in series and the oversizing factor as well as the rated apparent power of the inverter(s) and their rated power factor, the number of parallel modules is calculated internally. When the number of modules in series (parameter *n_series*) is adjusted, this has also an influence on the DC-voltage so that the minimal allowed DC-voltage in the *Controller* model (parameter *U_min*) might also needs to be adjusted. To access the *PV Array* model, open the composite model *Photovoltaic system* via the *plant model* link on the basic data page of the PV system:



The second row in the composite model contains the PV-array model (this is a model of the photovoltaic cell)

Changing the peak power has also an effect on the DC capacitance. Therefore, please have a look at the capacitance related to the active power (parameter C) in the “DC Busbar and Capacitor” model and adjust it, if required.

Changing the active power output

The active power is only defined by the PV system element data. If you would like to change the active power output you need to change this on the load flow page of the PV system element. The radiation is a result of the value entered on the load flow page of the PV system.

The active power output can also be limited during dynamic simulation via curtailment signal (pcurtail), which is based on the nominal active power of the PV plant. The active power limitation by the curtailement signal has priority over the active power reduction at overfrequency. Coversely, the active power reduction at overfrequency is applied if it does not conflict with the curtailment requirement.

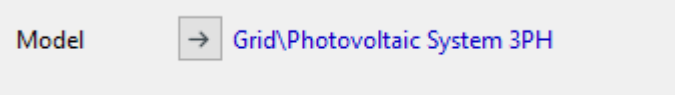
Changing the reactive power output

The following modes are available for the reactive power control:

- Reactive power setpoint
- Power factor setpoint, cos(φ)
- Reactive power characteristic as a function of the active power, Q(P)
- Reactive power voltage characteristic, Q(U)

- Reactive power with voltage limiting function

The settings for the reactive power control can be configured in the model *Qreference*. This model can be accessed by opening the composite model *Photovoltaic system* via the *plant model* link on the basic data page of the PV system:



The fifth row in the composite model contains the Qreference model (this is a model containing the reactive power control modes).

Changing the nominal AC voltage

You can also change the nominal voltage of the converter if your PV-system has a different nominal grid side voltage. The only modification you need to do is changing the nominal voltage of the terminal to which the PV system is connected.

Note: There is no saturation considered. It is assumed that the DC voltage is all the time high enough.

Changing the reactive current support during faults

The PV-model contains a reactive current support characteristic (Figure 4.1). This characteristic is activated during over- or undervoltage conditions and determines the amount of the additionally injected reactive current, linear to the voltage change. The parameters for the voltage thresholds regarding under and overvoltage condition for activating the fault mode and the parameter for the slope (droop) can be changed in the *Controller* model.

There is also the option available to enable the *Limited Dynamic AC Voltage Support Mode*, which is subsequently acitaved when the voltage drops below a user defined threshold. In this case, the maximum current that is injected is the maximum value specified by the user for this mode.

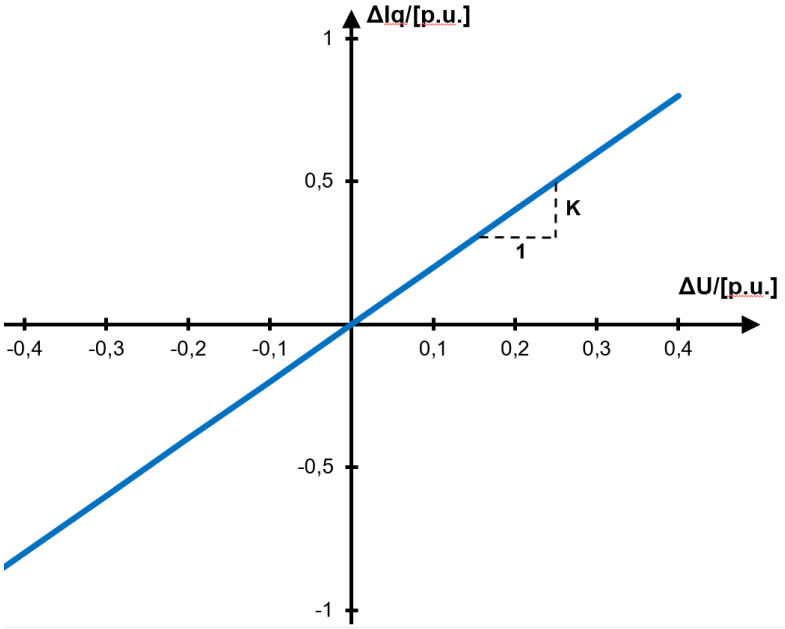
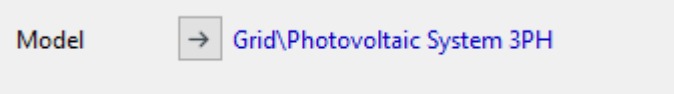


Figure 4.1: Reactive current support during voltage disturbances

1. Open the composite model *Photovoltaic System*. To find the correct composite model (if you have more than one) you can double click on the PV system in the single line diagram. On the basic data page is a link to the composite model:



2. Open in the composite model the *Controller* and make the changes according to your needs.

5 Using temperature and radiation profiles

The model supports also for dynamic simulation the usage of profiles (value over time) for the radiation and the temperature. The data, stored in a text file, has to be connected to *PowerFactory* via measurement file element (*ElmFile*). The format has to match the *PowerFactory* measurement file format, as described in [1].

The model comes already with two measurement file elements, connected to the slots “Measurement Solar Radiation” and “Measurement Temperature”. Connect the text file to one of the two elements, depending on the nature of the data. The unit of the temperature is °C and for radiation it is W/m^2. It is possible to store the data not only in the first data column but also in one of the other 24 data columns. It needs to be configured in the “Temperature” or in the “Solar Radiation” model which column should be used. Please note also that the measurement file elements as well as the previously mentioned selection models are by default out of service.

6 References

[1] *Technical Reference Measurement File*. Heinrich-Hertz-Strasse 9, 72810 Gomaringen / Germany, 2019.

[2] *Technical Reference PV System*. Heinrich-Hertz-Strasse 9, 72810 Gomaringen / Germany, 2019.

List of Figures

3.1 Frame Definition *Frame PV System* (*.BlkDef) 3

4.1 Reactive current support during voltage disturbances 6

List of Tables

3.1 Frame description 3